CE-116 (Revised)

2023

(April)

ENGINEEING GRAPHICS

(ECE & IT)

(1st Semester)

Full Marks : 30

Time : 3 hours

The Figures in the margin indicate full marks for the questions.

(Turn over)

1(a) A wheel 50 mm diameter rolls on a straight road surface without slip . Trace the path of a point which is at a radius of 30 mm from the centre of the wheel, for one complete revolution of the wheel .(10)

(b) A ball thrown from the ground level reaches a maximum height of 5m and travels a horizontal distance of 11m from the point of projection. Trace the path of the ball.(10)

2 (a) A regular pentagonal lamina of 30 mm base edges rests on one of its corners on H.P . Draw its projections when the surface of the plane makes 60° with the H.P and the top view of the diagonal passing through that corner on H.P makes 45° with the reference line. (10)

(b) State the quadrants in which the following points are located .(10)

(i) The projections of a point A coincide with each other at 50 mm below the xy line.

(ii) The front view of the point B is 40 mm below xy line and top view is 50 mm above the xy line

(iii) The front view of the point C is 30 mm below xy line and top view is 40 mm below the front view

(iv) The front view of the E is 40 mm below xy line and the top view is 30 mm behind the xy line.

3(a) The top view of a 75mm long line AB measures 65mm, while the length of its front view is 50mm. It's one end A is in the H.P and 12mm infront of the V.P. Draw the projections of AB and determine its inclinations with the H.P and the V.P.(10)

(b) A point Q is situated in first quadrant. It is 40 mm above H.P and 30 mm infront of V.P. Draw its projections and find its shortest distance from the intersection of H.P, V.P and auxiliary planes.(10)

4(a) A hexagonal pyramid , base 25 mm side and axis 50 mm long , has an edge of its base on the ground . Its axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections.(12)

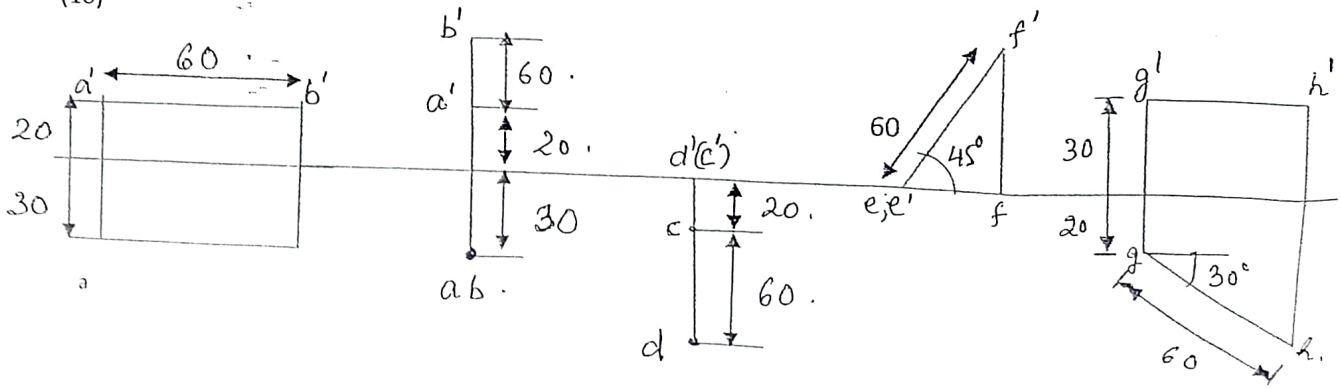
(b) A 100mm long line is parallel is parallel to and 40mm above H.P. Its two ends are 25 mm and 50 mm infront of the V.P respectively. Draw its projections and find its inclinations with the V.P. (8)

5(a) Draw a straight line AB of any length . Mark a point F, 65mm from AB. Trace the paths of a point P moving in such a way that the ratio of its distance from the point F, to its distance from AB is 3:2. Plot atleast 8 points . Name the curve. Draw a normal and a tangent to the curve at a point on it, 50 mm from F. (12)

(b) A point P is 20mm below H.P and lies in the third quadrants . Its shortest distance from xy is 40 mm . Draw its projections. (8)

6(a) Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the H.P and inclined at 60° to the V.P., and its surface making an angle of 45° with the H.P. (10)

(b) State the positions of the straight lines with reference to V.P and H.P of their projections are as in (10)

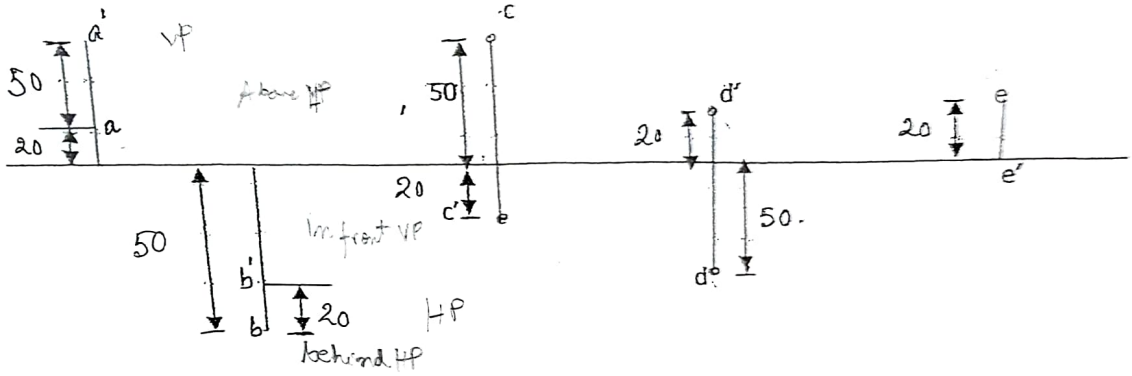


7(a) The front view of the line AB of length 70 mm is inclined at 30° to the xy line and measures 45mm. The end A is 20mm above H.P and 25mm in front of V.P. draw the projections of the line and find the inclinations with the H.P and V.P.(10)

(b) A circle of 50mm diameter rolls along a straight line without slipping. Draw the curve traced out by a point P on the circumference, for one complete revolution of the circle. Name the curve. Draw a tangent to the curve at a point on it 40mm from the line. (10)

8(a) Draw the involutes of a regular hexagon of side 20mm and mark a tangent and a normal to the curve at a distance 10mm from the centre of the hexagon.(10)

(b) The orthographic projections of various points are shown in figure. State the position of each point with respect to the reference planes.(10)



9(a) A line AB of length 70mm is inclined at 30° with H.P and 45° with V.P. The end A is 20mm above H.P and 30mm infront of V.P. Draw the projections of the line and finds its apparent inclinations with respect to the principal planes. (10)

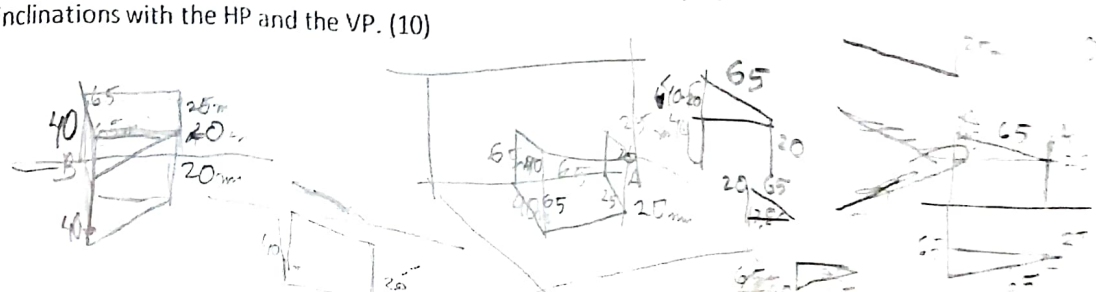
(b) A regular hexagonal lamina of 30mm sides rests on H.P on one of its sides. The side which is on H.P makes 60° to the V.P and the surface of the lamina is inclined to H.P at 45°. Draw the front view and top view of the lamina in its final position. (10)

10(a) A pentagonal prism 30mm side of base and axis 70mm long is resting on one of its edges of the base, in such a way that the base makes an angle of 40° with H.P. draw the projections if the axis is parallel to V.P.(12)

(b) A circular lamina of 60mm diameter rests on H.P such that the surface of the lamina is inclined at 30° with H.P and perpendicular to V.P. Obtain its projections. (8)

11(a) A circle of 50mm diameter rolls on the circumference of another circle of 175mm diameter and outside it. Trace the locus of a point on the circumference of the rolling circle for one complete revolution. Name the curve. Draw a tangent and a normal to the curve at a point 125mm from the centre of the directing circle. (10)

(b) A line AB, 65mm long, has its end A 20mm above the H.P and 25mm in front of the V.P. The end B is 40mm above the H.P and 65mm infront of the V.P. Draw the projections of AB and shows its inclinations with the HP and the VP. (10)



12(a) A wheel 50mm diameter rolls on a straight road surface without any slip. Trace the path of a point on one of the spokes 10mm from the rim towards the centre of the wheel for one complete revolution. (10)

(b) Draw the projections of a regular hexagon of 25mm side, having one of its sides in the HP and inclined at 60^0 to the VP and its surface making an angle of 45^0 with the HP. (10)

13(a) Draw the projections of a pentagonal prism, base 25mm side and axis 50mm long, resting on one of its rectangular faces on the HP, with the axis inclined at 45^0 to the VP (12)

(b) A vertical line AB, 75mm long, has its end A in the HP and 25mm in front of the VP. A line AC, 100mm long, is in the HP and parallel to the VP. Draw the projection of the line joining B and C, and determine its inclinations with the HP. (8)

14. Draw the isometric projection of a pentagonal prism of 22mm base edges and axis 50 mm long when it has one of its rectangular faces on HP and axis perpendicular to the VP. (20)